

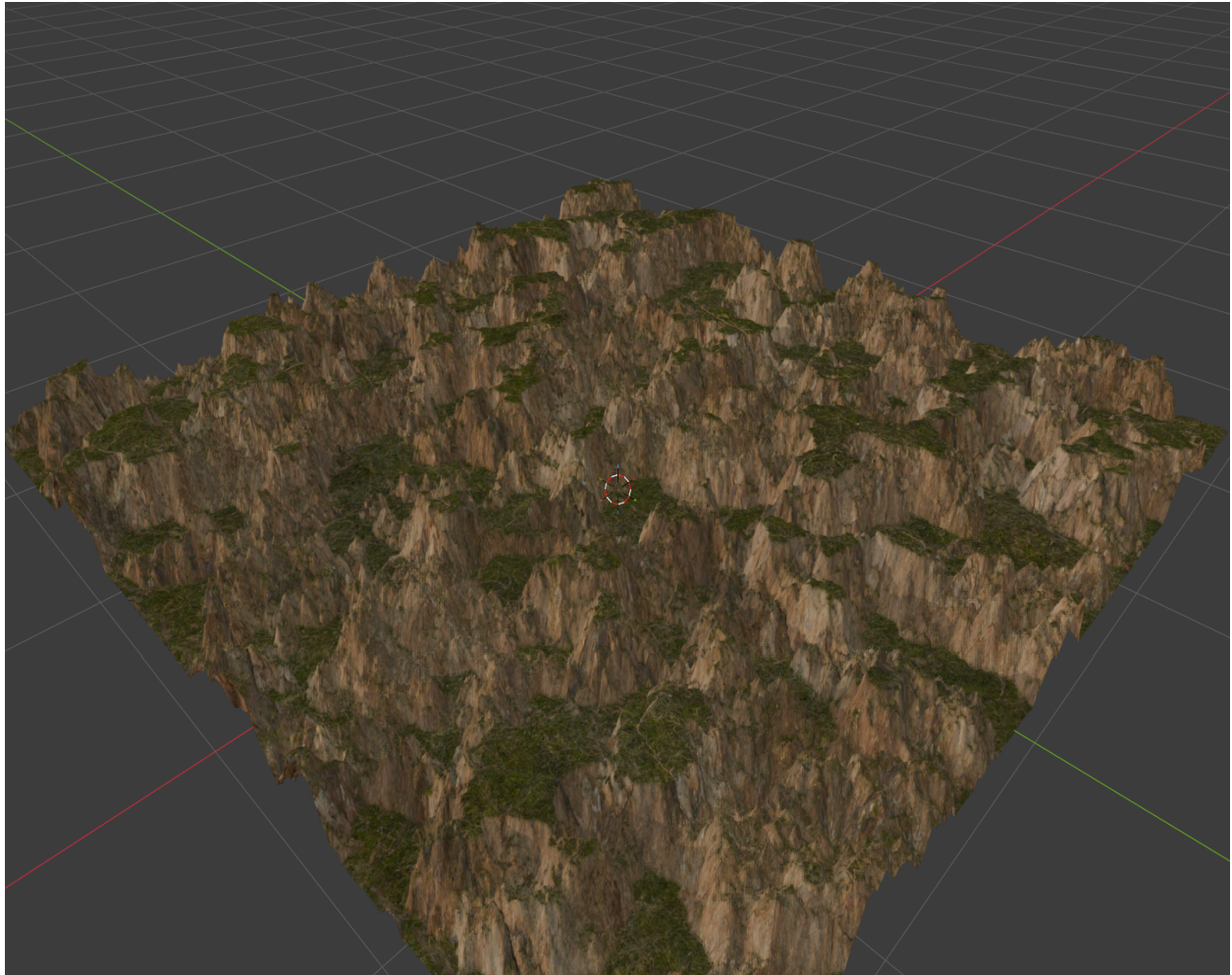
Infinite World Generator
Project Report
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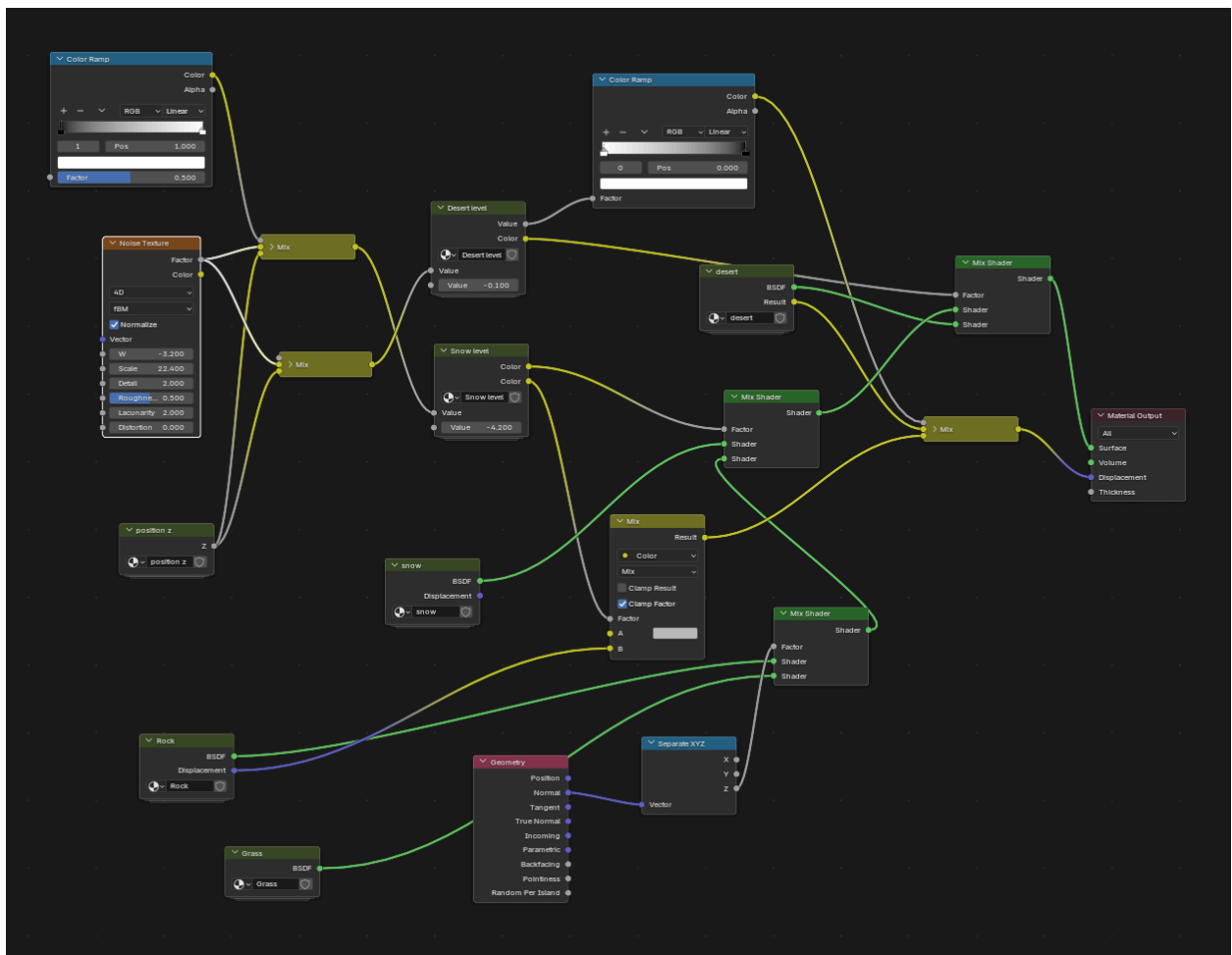
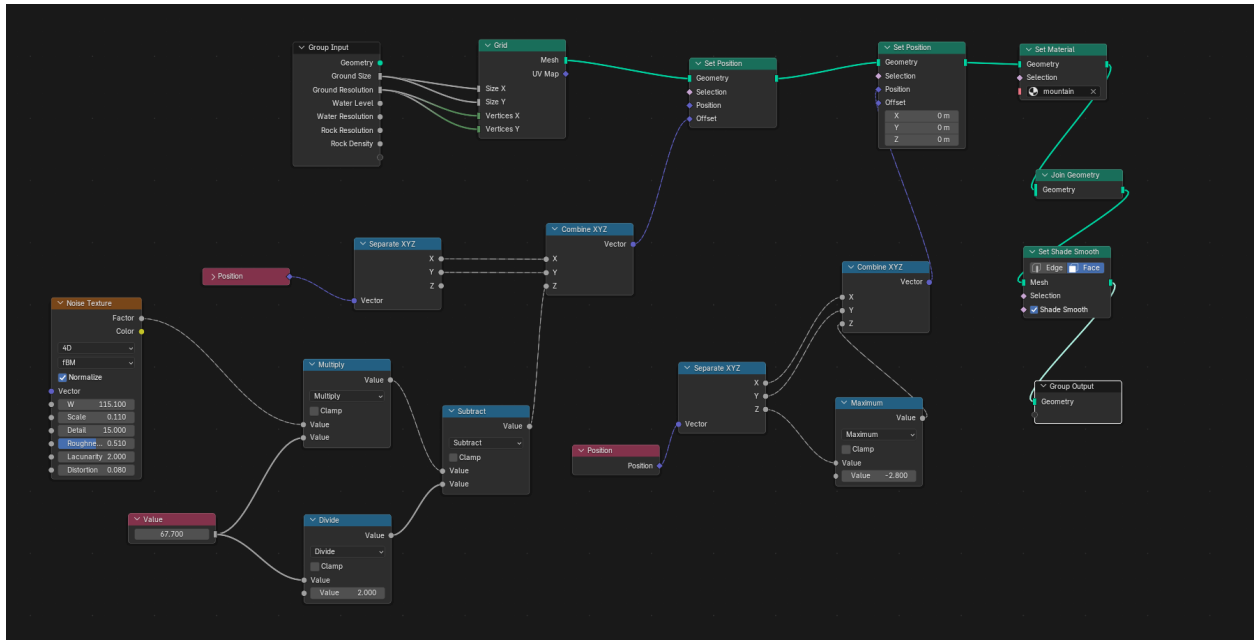
Week 1

For the first week, the goal was simply to use Blender and Geometry Nodes to create simple terrains procedurally.

For this, I made mountain-like structures with hills and canyons.

The node logic for this one was simple. I offset a grid using two noise textures. Two noise textures were used for fine-tuning the result. The colour ramp was used to control the height of the mountain and the base. And I imported some textures to add a rock + grass look to the mountain.

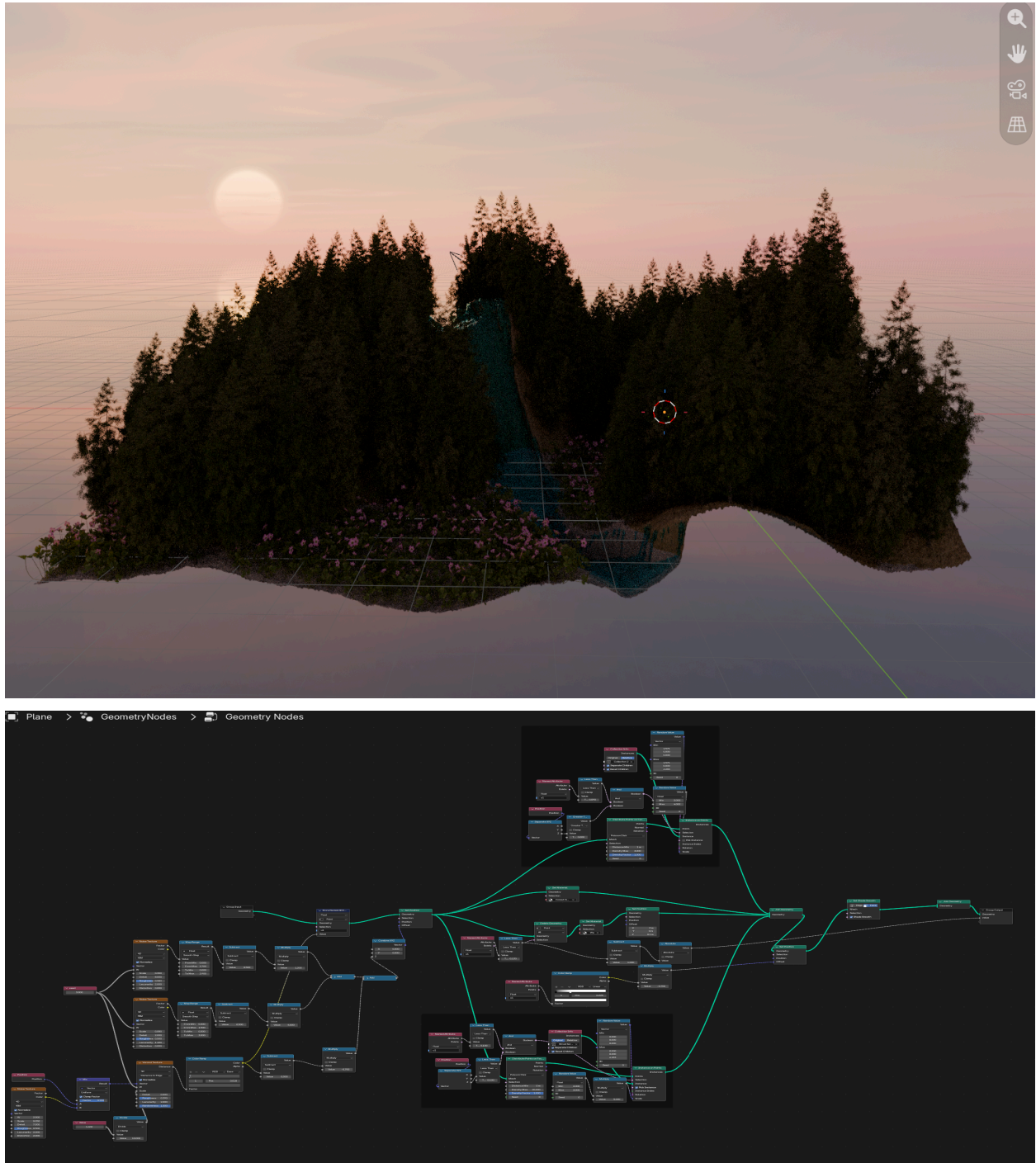




Week 3

For week three, the goal was to make a river and a forest.

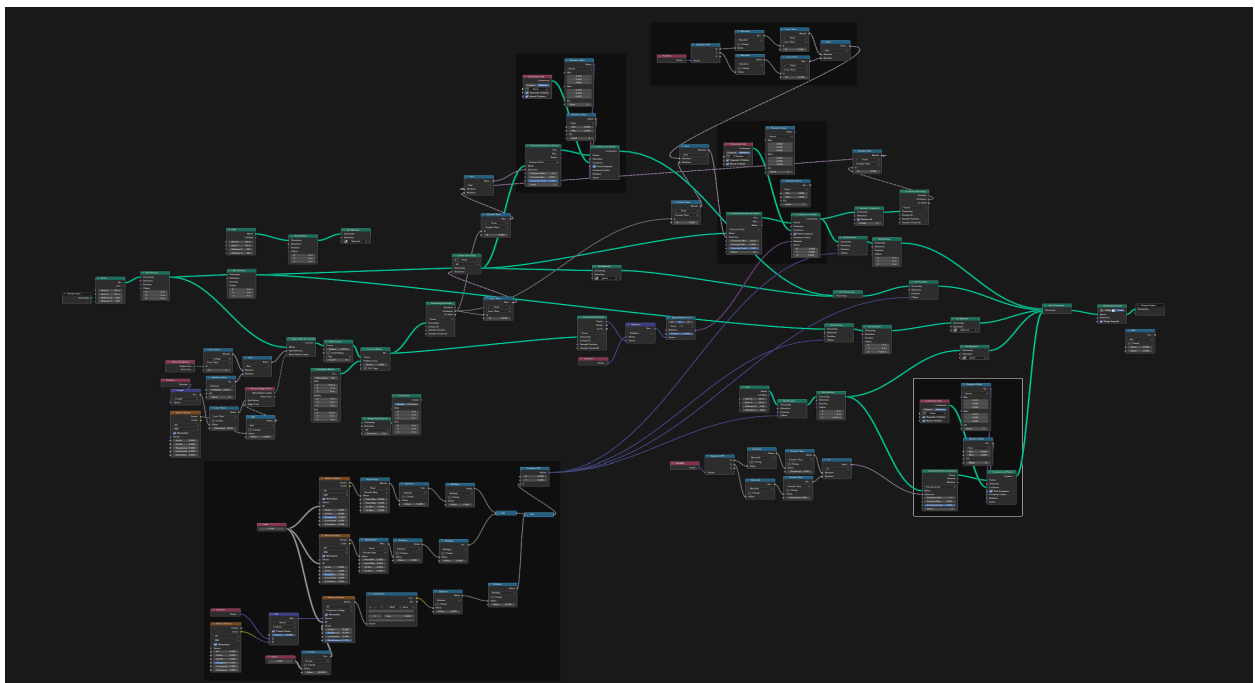
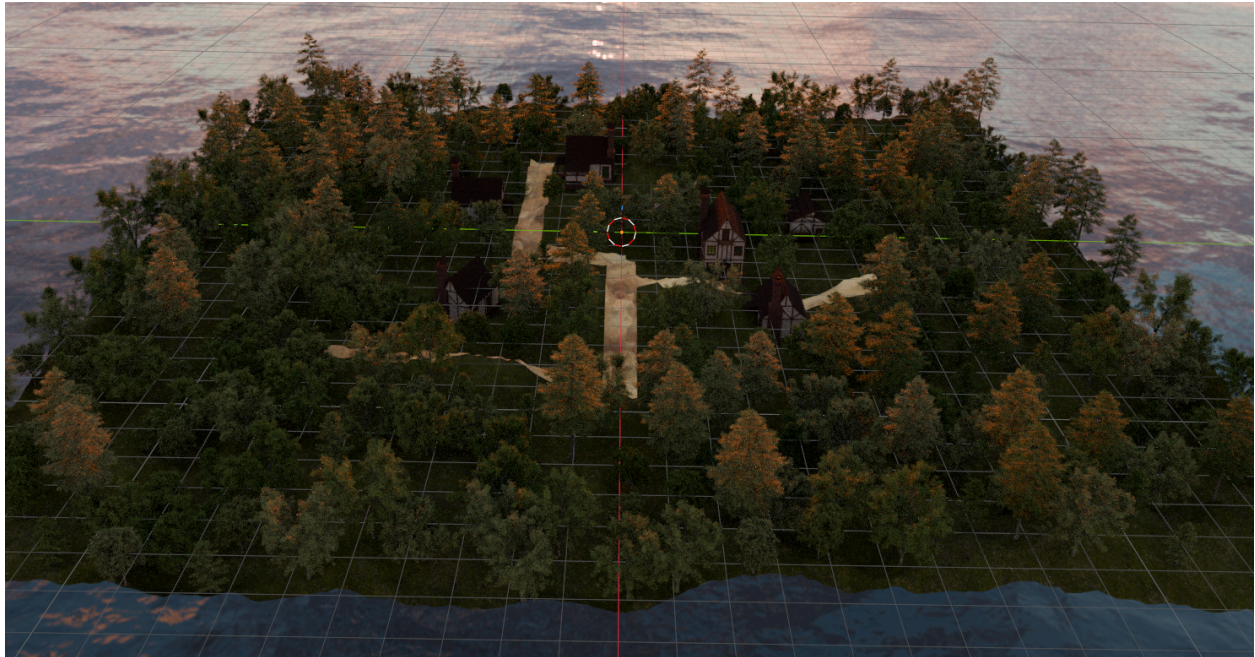
For this, on a noisy, hilly environment, I used a Voronoi texture to map out the river, then stored it as a named attribute. This named attribute was used to apply material to the river and to ensure the vegetation does not spawn on the river. Finally, using point distribution and object distribution functions, I placed trees and flowers on the remaining terrain.



Week 4

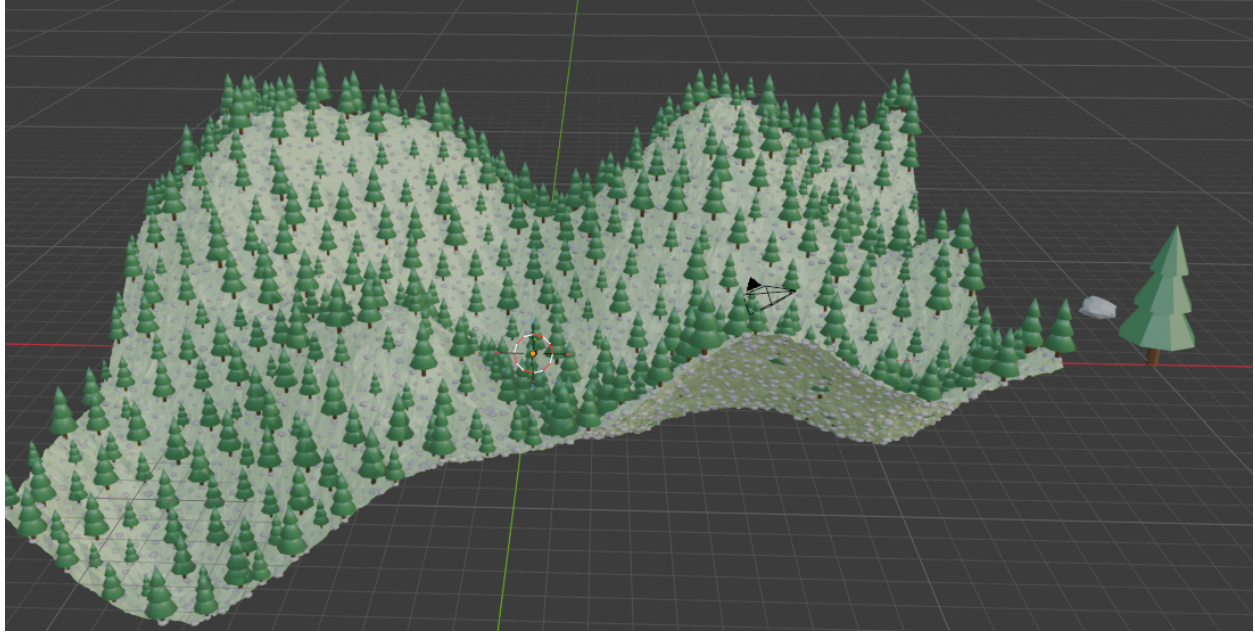
For week 4, the goal was to introduce settlements. I made a small forest settlement, with roads and houses in the middle of a forest.

For this, I used a smaller grid inside a larger grid as a perimeter for my settlement. Then using vertex selection and path generation function nodes, I was able to make a random path inside the perimeter. Then excluding the area of the path, I placed houses and trees. Geometry proximity nodes and a solid logical chain was used to ensure that the trees and houses don't spawn at the same place. The area outside the perimeter was then covered with trees. Finally, I imported some textures and backgrounds to create the final render.



Week 5

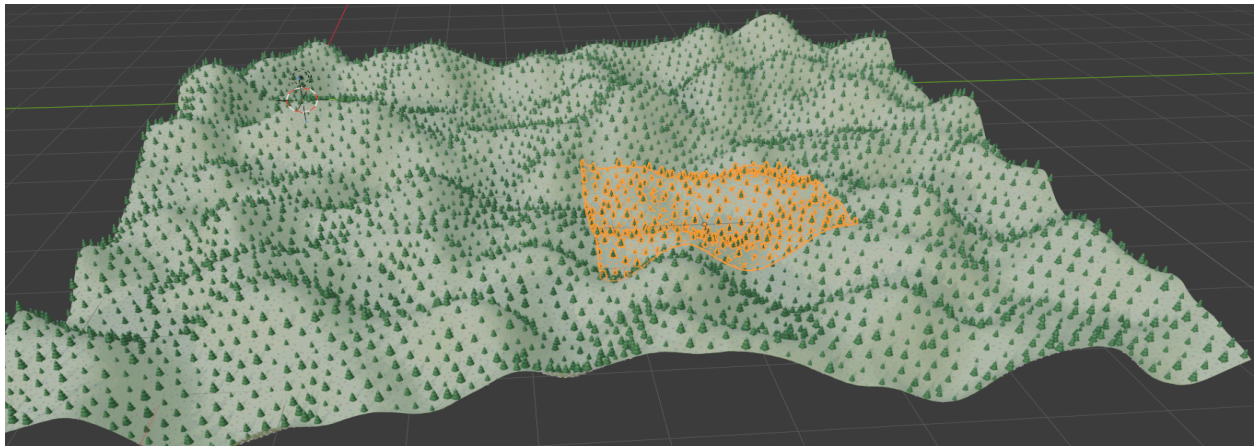
For week 5, the goal was to transition from visual nodes to Python scripting inside Blender. For this, I used the Blender Python API to access and control my scene data using the script. Instead of manually creating and adjusting nodes, I wrote a script to initialize nodes, set their parameters and link them. I also made a procedural low poly tree and rock using scripts. The final script is was able to generate a hilly plane with vegetation. All parameters like size of the plane, master seed were hardcoded in the script.



Week 6

For week 6, the goal was to construct the underlying mathematical algorithms capable of generating infinite, reproducible landscapes.

For this, I coded a chunk-based generation pipeline in Python. By treating the world as an infinite 2D coordinate grid, the script calculates terrain data based on specific chunk coordinates and a master seed. This coordinate math is then passed directly into the geometry nodes modifier to offset the chunks seamlessly, while the raw generated data is saved to a JSON file for storage.



Week 7

For week 7, to create a final blender ad-on built on the node logic and the scripts from previous weeks.

For this, I designed and registered a custom UI panel in Blender's sidebar. I exposed all the necessary world parameters—like Grid Size, Subdivisions, Mountain Strength, and Snow Level—and connected them to the procedural backend using a parameter dictionary. This combined the entire project into the final "Infinite World Generator" add-on, allowing users to generate entirely new worlds interactively.

